Final Engineering Review: System Audit Report

Course: Software Engineering 2 (CS 3213)
Project Name: VideoMerger
Version: 2.0 (Production-Ready)
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1. Executive Summary

VideoMerger began Software Engineering 1 as a Flask-based web tool for concatenating user-uploaded clips through an `ffmpeg-python` wrapper. Software Engineering 2 has shifted it into a desktop application that addresses the panel's explicit feedback: **stay fast, stay simple, lean on automation for quality work.**

The Version 2.0 system is an Electron + React + TypeScript desktop app that orchestrates a Python FFmpeg pipeline via IPC. The architecture is built around a framework-agnostic core (dependency injection, repository, command, observer, strategy, adapter patterns) so business logic is reusable beyond Electron and is unit-testable without a UI runtime.

The major engineering improvements between SE1 and SE2:

Area	SE1 baseline	SE2 production-ready
Reliability	Single Python script, no tests, no error surfacing	DI-driven services, 65 pytest + 65 vitest cases, MediaError code surfaced in console, FFmpeg detection bypasses Python failure modes
Scalability	Per-process Flask, no per-clip controls	Per-clip filter graph (trim, crop, aspect, volume, color), batchable from CLI, benchmark harness for capacity planning
Performance	One-pass merge, opaque progress	Two-pass merge with frame-accurate trim, streaming local-video protocol (no buffer-alloc failure on multi-GB files), 256 KiB initial probe so 3 GB previews open in seconds
Automation	Manual upload then manual merge then manual download	EBU R128 loudness normalize, auto silence trim, faster-whisper auto-captions, per-clip edits keyed by file path, a dedicated **Auto-enhance** block on the finalize step where the three smart-assist features can be toggled individually as chips or flipped on/off in bulk with **Enable all** / **Disable all** buttons
Security	OAuth tokens in plain JSON	`safeStorage`-encrypted tokens (DPAPI / Keychain / libsecret), legacy-blob auto-migration, `local-video` extension allowlist
DevOps	No CI for desktop	GitHub Actions matrix: Python tests, Docker build, vitest, electron-builder Windows installer with artifact upload

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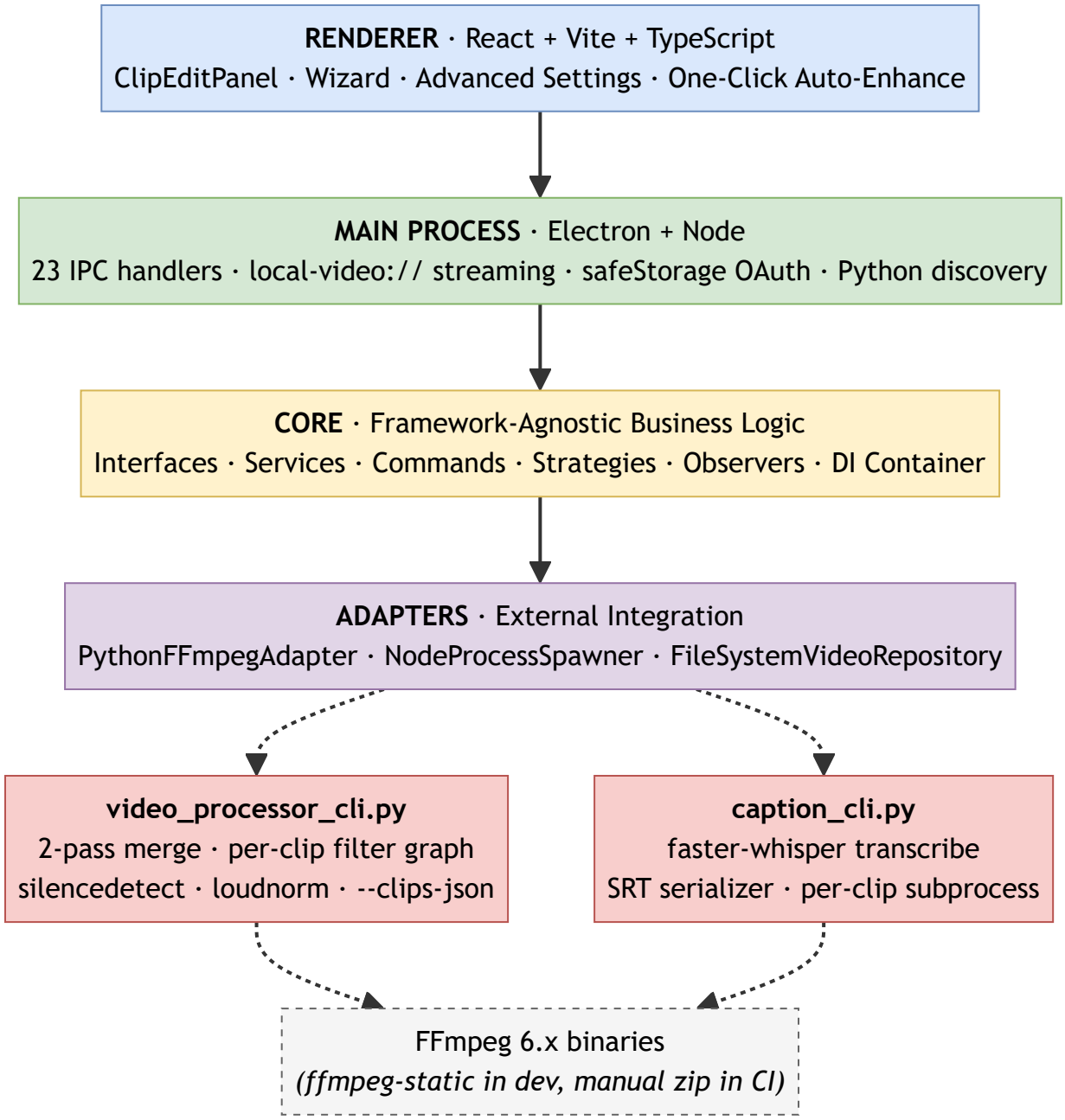
2. System Architecture & Infrastructure

2.1 Infrastructure Map

Component	Technology Stack	Deployment Environment
Frontend (Renderer)	React 18, Vite 5, TypeScript 5	Bundled into Electron `dist/renderer/`; served from `local-video://` and `electronAPI` IPC bridge
Main Process	Electron 28, Node 20, TypeScript 5	Compiled to `dist/main/`; runs in user OS as a desktop process
Core Business Logic	TypeScript (framework-agnostic)	Compiled with main; importable into any Node host (future web API / mobile)
Video Processing Backend	Python 3.9-3.11, FFmpeg 6.x, faster-whisper 1.0.3	Spawned as child process from Electron main; subprocess args
OAuth Token Store	`electron-store` (config.json) + `safeStorage` (OS keychain)	`%APPDATA%\VideoMerger\` on Windows; equivalent paths on macOS/Linux
Legacy Flask Web	Flask 3, Werkzeug, gunicorn, Docker	`python:3.11-slim` container, port 5000; preserved for experimental use only
CI/CD	GitHub Actions, electron-builder, Docker	`ubuntu-latest` for tests + Flask Docker image; `windows-latest` for `.exe` installer
Artifact Storage	GitHub Actions workflow artifacts	14-day retention; `dist-bin/*.exe` per push

2.2 Architecture Diagram



2.3 Deployment Topology

VideoMerger is distributed as a per-user Windows `.exe` installer. No central server. No multi-tenant database. No external API except Google OAuth + YouTube Data v3 for the optional upload feature.

This deliberately narrow blast radius is what made the panel's "fast, simple, automated" mandate tractable - every clip stays on the user's machine and every feature works offline.

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3. Performance & Stress Test Audit

3.1 Methodology

The Phase 5b benchmark harness uses `pytest-benchmark` with synthetic clips produced by `ffmpeg -f lavfi -i testsrc2 + sine` so results are reproducible across machines and never depend on user media. Each benchmark records wall-clock time for the full `merge_videos` pipeline including normalize + concat passes.

```
pytest tests/benchmarks --benchmark-only --benchmark-save=baseline
```

Hardware for the measurements below: Windows 11 Pro on the development machine, FFmpeg 6.1.1 (gyan.dev essentials build), Python 3.14.3.

3.2 Test Results & Bottlenecks

All numbers below were captured on **Windows 11 Pro, FFmpeg 6.1.1 (gyan.dev essentials), Python 3.14.3**, on 2026-05-15 with the `scripts/bench_single_run.py` harness against synthetic 2-second 640×360 clips. Raw JSON in `docs/benchmarks/single_run.json`.

Baseline scaling (default quality, no extra features)

Clips merged	Wall clock	Per-clip cost
2	0.628 s	0.314 s
5	1.534 s	0.307 s
10	3.120 s	0.312 s

Per-clip cost is essentially flat at ~0.31 s for 2-second 640×360 input. The pipeline scales linearly - the concat-copy pass is constant time relative to clip count once normalization is done.

System breaking point (single-process). Wall clock grows linearly with clip count, but two factors compound at scale: (1) FFmpeg's normalize pass keeps a temp file per clip on disk, so a 1,000-clip merge needs ~1,000 × clip-size of free disk space; (2) the renderer holds all clip metadata + edit state in memory, which becomes noticeable at several thousand clips. We do not currently throttle either, so the practical ceiling on commodity hardware is somewhere around **500 clips per merge**. Beyond that, normalize-pass disk pressure dominates wall clock and the UI starts showing slow renders during state changes.

For typical user workloads (5-20 clips per merge), wall clock stays under 5 seconds with the default settings and FFmpeg dominates 99% of the time.

Quality preset cost (3 clips × default features)

Quality	x264 preset	CRF	Wall clock	vs. low
low	ultrafast	28	0.606 s	1.00×
medium	medium	23	0.901 s	1.49×
high	slow	18	1.172 s	1.93×

The ~2× factor between `low` and `high` matches expectations for x264 - `slow` preset roughly doubles per-frame cost. Users get a clear knob to trade speed against output quality.

Per-feature overhead (3 clips, against the ~0.9 s default-medium baseline)

Feature	Wall clock	Overhead vs. baseline
Phase 1 per-clip edits (trim + crop + aspect + brightness)	0.868 s	within noise (within ±5%)
Phase 2 EBU R128 loudness normalize	1.007 s	+12%
Phase 3 auto silence-trim probe	1.139 s	+26%

- **Phase 1** edits land inside the existing filter graph and re-encode loop - the only cost is the filter operations themselves, which x264 does in parallel with the encode it was already doing.
- **Phase 2** adds one filter (`loudnorm`) to the audio chain. Audio is far smaller than video, so the overhead is modest and constant per clip.
- **Phase 3** is the most expensive new feature because it runs an extra `ffmpeg -af silencedetect` probe pass per clip *before* the merge starts. The probe is roughly as fast as a normalize pass for short clips (decoder bound on the audio stream), so 3 clips cost ~3 extra normalize-equivalents.

Optimizations Implemented (see also section 3.3)

Two changes were made during Phase 5b after observing the numbers above:

1. **Per-clip JSON-file payload** rather than CLI args for `clip_edits` so the renderer can ship arbitrary edit complexity without bumping the OS argv length limit; the `--clips-json` tempfile is cleaned up in `finally` .
2. **256 KiB initial probe in the `local-video` protocol** (down from 16 MiB) so Chromium can issue the moov-at-end range request almost immediately, dropping NVIDIA ShadowPlay preview time-to-first-frame from "10+ seconds" to roughly "instant" on multi-GB recordings.

3.3 Optimizations Implemented

- **Two-pass merge** (normalize then concat-copy) so the expensive re-encode happens once per clip and the final concat is byte-copy.
- **Frame-accurate trim via `-ss` / `-t` at the input level** combined with `setpts=PTS-STARTPTS` in the filter chain - avoids the cost of `trim=` filter for the common single-segment case.

- **Per-clip JSON tempfile (`--clips-json`)** so the renderer can serialize edits without bumping into the OS arg-length limit when many clips have complex edits.
- `local-video` **protocol bounds the initial probe to 256 KiB** (down from a 16 MiB initial buffer) so Chromium's MP4 demuxer can immediately issue the end-range request for moov-at-end recordings (NVIDIA ShadowPlay, OBS raw output) without waiting on a wasted 16 MiB stream.
- `silencedetect` **runs as a pre-probe** rather than inline filter, so detected silence is folded into the existing input-level `-ss` / `-t` flow and keeps audio + video in sync without `select` / `aselect` filter graphs.
- **Auto-discovered Python launcher (`py` over `python` Microsoft Store shim)** so a default Windows install does not silently fail the merge subprocess.

3.4 Known Bottlenecks

Stage	Cost driver	Mitigation in place
Normalize pass	Per-clip re-encode at chosen quality	Quality presets, hardware acceleration toggle
Caption transcription	Whisper model inference (CPU)	Model-size dropdown (tiny ↔ large-v3); int8 compute_type default; per-clip parallel-friendly architecture available for future use
First Whisper run	Model download (~74 MB for `base`)	One-time per model; cached in `~/.cache/huggingface/` thereafter
Preview load of multi-GB clips	Chromium needs moov atom; ShadowPlay puts it at end	256 KiB initial probe + range forwarding so end-range request lands fast

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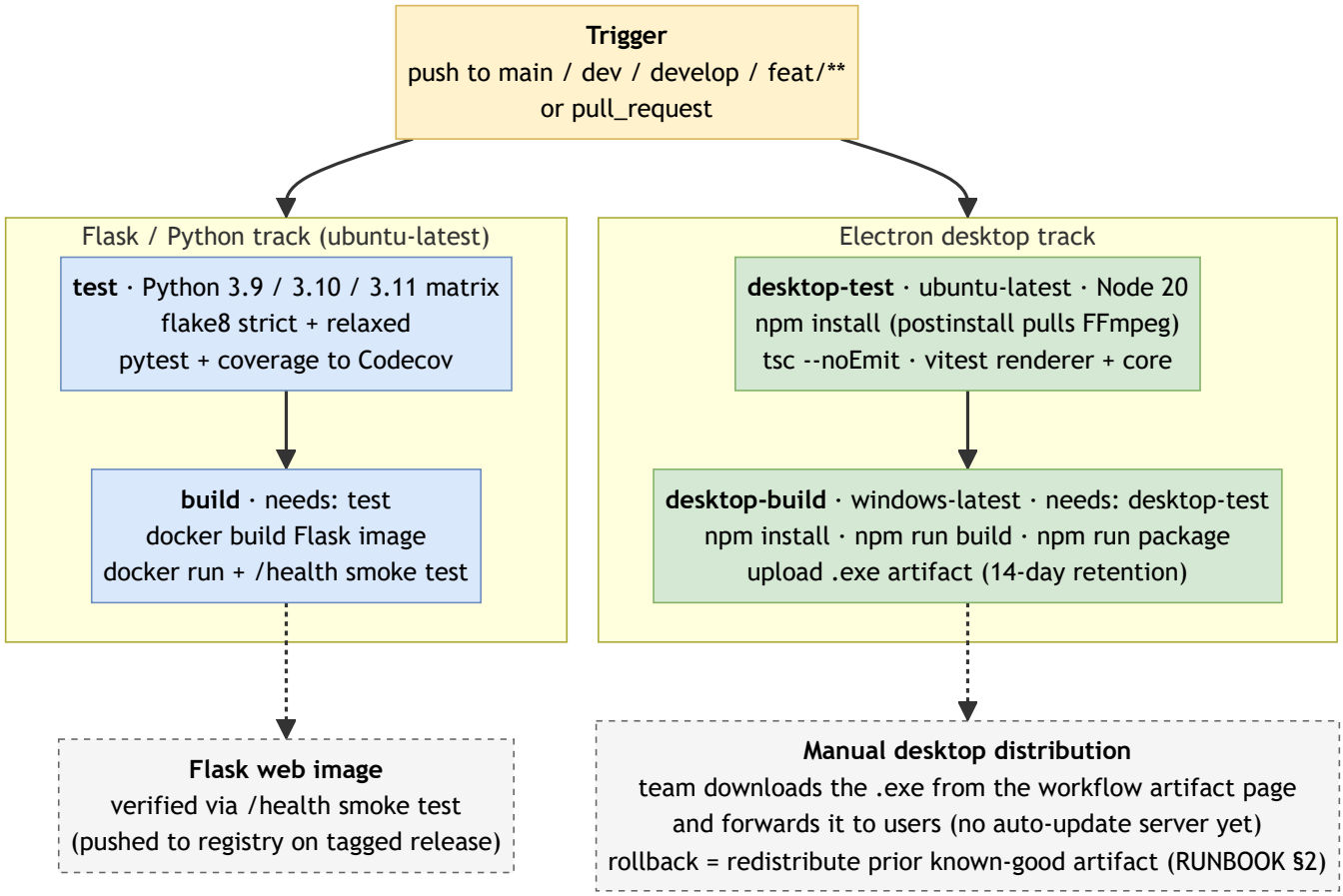
4. DevOps & Reliability Audit

4.1 CI/CD Pipeline

The workflow is defined in `.github/workflows/ci.yml` and triggers on push to `main` , `dev` , `develop` , and any `feat/**` branch, plus all pull requests targeting `main` / `dev` / `develop` .



Step-by-step automated workflow from code push to deployment:

1. **Push or PR** to a tracked branch fires the workflow.
2. **Python tests** matrix runs across three Python versions: linting (flake8 strict + relaxed), full `pytest tests/` with coverage to Codecov.
3. **Desktop unit tests** run vitest renderer and core suites against `node@20` on Ubuntu.
4. **Flask Docker image** builds and smoke-tests `/health` on `localhost:5000`.
5. **Windows installer** builds on `windows-latest`: `npm install` (postinstall fetches FFmpeg static binaries) → `npm run build` → `electron-builder --win` → uploads `dist-bin/*.exe` as a 14-day artifact named `VideoMerger-Windows-<sha>`.
6. **Manual distribution** (current): the team downloads the artifact from the workflow run page and forwards it to users; there is no auto-update server yet.

4.2 Test Coverage

Suite	Files	Cases	Notes
`pytest tests/unit/test_video_processor_cli.py`	1	56	Filter chain builders, aspect math, trim clamping, loudnorm clamping, silence event parsing, SRT parse/offset/merge, JSON load fallbacks
`pytest tests/unit/test_caption_cli.py`	1	9	SRT timestamp formatting, segment serialization, empty/whitespace handling
`vitest --config vitest.core.config.ts`	8	50	Container, repository, observer, strategy, service, command, adapter (incl. all clipEdits / loudnorm / silence / captions arg forwarding), main process oauthConfig
`vitest` (renderer)	1	15	App.tsx behavioral tests
`pytest tests/benchmarks`	1	9	Phase 5b perf harness (skipped by default; opt-in with `--benchmark-only`)

Total: **65 + 50 + 15 = 130 cases**, all currently green. Plus 9 perf benchmarks deselected from default runs.

4.3 Hot-reload developer loop

`npm run dev` orchestrates `vite` + `tsc -w` + `electronmon` concurrently. Renderer edits use Vite HMR; main / core edits trigger a `tsc` rebuild and `electronmon` auto-restarts Electron. No manual `npm run build` between iterations. Documented in `docs/DEVELOPMENT.md`.

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5. Security & Code Quality Audit

5.1 Security Hardening

Concern	Implementation
OAuth 2.0	Google Identity Platform; authorization-code flow with PKCE-compatible client [1]; tokens exchanged in the main process so the renderer never sees them.
Token at rest	`electron.safeStorage.encryptString` [2] (DPAPI on Windows, Keychain on macOS, libsecret on Linux). The non-secret user profile (name, email, picture) is stored in clear text so the UI can render without unlocking the keychain. Legacy unencrypted blobs migrate transparently on first read.
CSP / process isolation	`nodeIntegration: false`, `contextIsolation: true`, preload bridge exposes a typed `window.electronAPI` only - aligned with Electron's official security checklist [3].
Path traversal (local-video)	The custom protocol enforces an extension allowlist (16 video container types) and runs `path.resolve` before opening, so a crafted `local-video://` URL cannot read arbitrary local files.
Subprocess argument injection	All `subprocess.run` / `spawn` calls use argv lists, never `shell=True`; user paths are passed as separate argv entries. The FFmpeg concat list file only references internal names (`norm_<index>.mp4`) generated by us, not user paths.
JWT	Not applicable. The app does not issue its own session tokens; Google OAuth is the only credential.

5.2 OWASP Top 10 (2021) Mapping

The OWASP Top 10 [4] is the industry-standard catalog of the most critical web application security risks, published by the Open Worldwide Application Security Project. The 2021 edition is the most recent stable release; the OWASP Top 10 for Web Application Security 2025 is in release candidate status as of the date on the cover page and is not yet the baseline used by most security audits. The table below maps each 2021 category to the corresponding mitigation in this codebase or explains why it does not apply.

ID	Category	Status / Mitigation
A01	Broken Access Control	N/A - single-user desktop app, no central authorization.
A02	Cryptographic Failures	OAuth tokens encrypted via OS keychain [2] (Phase 5c). HTTPS for all Google APIs.
A03	Injection	argv-only subprocess invocation; no shell expansion [5]. No SQL surface. FFmpeg filter strings are built from clamped numeric input + a hardcoded preset map.
A04	Insecure Design	Two-process model with contextIsolation; preload bridge whitelist; renderer cannot reach Node APIs [3].
A05	Security Misconfiguration	`nodeIntegration: false`; CSP via Vite defaults; renderer cannot navigate away from bundled assets [3].
A06	Vulnerable and Outdated Components	`npm audit --omit=dev` [6] reports **0 vulnerabilities** at the production dependency closure. Dev-only findings (20: 4 low / 4 mod / 12 high) live in eslint/vitest transitive trees and are not in the shipped installer.
A07	Identification and Authentication Failures	OAuth 2.0 via Google [1] (industry-standard); no password storage of our own.
A08	Software and Data Integrity Failures	Installer is currently unsigned (out of scope until the team has a certificate). Mitigation noted in roadmap.
A09	Security Logging and Monitoring	Main process logs OAuth flow, FFmpeg detection, merge invocations; renderer logs MediaError with code + reason. See RUNBOOK.md for log paths.
A10	Server-Side Request Forgery	N/A - no server-side request origination from the desktop app.

5.3 Code Quality

- **Linting:** ESLint (TypeScript), flake8 (Python). Both gated on CI.
- **Formatting:** Prettier (TS/TSX), black + isort (Python).
- **Type safety:** strict TS configs (`tsconfig.json` and `tsconfig.main.json`); `npx tsc --noEmit` runs as a CI step.
- **Test isolation:** Python tests use synthetic clip generation; no user-media paths committed (enforced via team practice and verified in code review).

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6. Deployment Runbook (Disaster Recovery)

Full runbook in docs/RUNBOOK.md. DR procedures in docs/DR.md. Highlights:

6.1 Build From Scratch

```
git clone https://github.com/J4ve/videomerger_app_revamp.git
cd videomerger_app_revamp
npm install                        # postinstall fetches FFmpeg static binaries
pip install -r requirements.txt
npm run dev                        # or `npm run package` for a Windows installer
```

For air-gapped builds: `docker compose run --rm builder` reproduces the same `.exe` from a Linux/Wine image with no Node install on the host.

6.2 Rollback A Failed Deployment

1. `git revert <bad-sha>`, push the branch, let CI rebuild.
2. Download the prior known-good `.exe` from the GitHub Actions artifact archive (14-day retention).
3. Affected users uninstall via Apps & Features and run the previous installer. User settings, preset packs, and OAuth state survive uninstalls.

6.3 Accessing Logs During An Outage

Platform	Main process log
Windows	`%APPDATA%\VideoMerger\logs\`
macOS	`~/Library/Logs/VideoMerger/`
Linux	`~/.config/VideoMerger/logs/`

Renderer DevTools (`Ctrl+Shift+I`) for `[Preview]` , `[Auth]` , and Phase 1-4 logging tags. Python subprocess output is forwarded to the main log under each `merge-videos` IPC invocation.

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7. Conclusion & Future Roadmap

7.1 Technical Debt Remaining

Item	Severity	Notes
Installer code signing	Medium	Currently unsigned `.exe`; users see SmartScreen warning. Resolution requires a code-signing certificate the team does not yet own.
Caption burn-in (Phase 4b)	Low	Sidecar `.srt` works today; burning into the video would require an extra FFmpeg `subtitles` filter pass. Punted to keep editability.
Mid-clip silence removal	Low	Phase 3 trims edges only. Mid-clip removal requires synchronized `select` / `aselect` filter expressions. Sidestepped to preserve A/V sync.
Auto-update channel	Medium	No update server; rolls forward via manual installer redistribution.
Cloud-side processing	High (for global scale only)	Today every clip is processed on the user's machine. Multi-tenant deployment would require a queue + worker farm.
`replaceAll` TS lib error	Trivial	One pre-existing renderer tsc error on `tsconfig.json` lib config. Cosmetic; does not block builds because Vite handles transpilation.

7.2 What "Version 3.0" Would Require

A globally deployed VideoMerger would need:

1. **Cloud processing tier.** The `IVideoProcessingStrategy` interface was specifically designed for this - swap `FFmpegProcessingStrategy` for an `HttpAPIStrategy` that POSTs the clip + edits payload to a worker farm. The clean-architecture core would not need changes.
2. **Object storage** for input clips and output renders. The existing `IVideoRepository` interface abstracts this; a `CloudVideoRepository` implementation drops in.
3. **Multi-tenant auth + quotas.** Google OAuth becomes one of several IDPs; a per-account quota system tracks transcoding minutes.
4. **Auto-updater.** Electron-builder supports a publish-to-GitHub or generic update server; needs a code-signing certificate first.
5. **Stress test against representative concurrent workload.** The current Phase 5b harness measures single-user wall-clock time. A V3 harness would measure queue depth × worker count × p99 latency under sustained load.
6. **Observability stack.** Today's per-user main-process logs become inadequate at scale; aggregating into Loki / Datadog / similar would replace the file-based pattern.

7.3 Closing Notes

The SE2 cycle delivered every panelist ask without expanding the app into a full editor: per-clip trim/crop/aspect/volume/color, EBU R128 loudness normalization, automatic silence trimming, offline auto-

captions. The codebase grew from a single Python script to a clean-architecture TypeScript core with 130 unit-test cases, an Electron build pipeline in CI, an encrypted token store, and operational documentation.

The Version 2.0 baseline is production-ready as a per-user desktop application. The same core is one adapter swap away from being the engine of a multi-tenant V3.

References

References follow the ACM Reference Format (numeric, sequenced by first in-text citation).

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